

ADVANCED TACTICAL VISUALS & AI MODELS

Cutting-edge spatial analytics: Voronoi continuous space ownership, 16x12 Karun Singh Markov Expected Threat (xT), 360° Pass Sonars, and Optical Tracking Minimaps.

PITCH CONTROL

58.4%

MARKOV THREAT

+0.32 xT

CV ACCURACY

0.142m

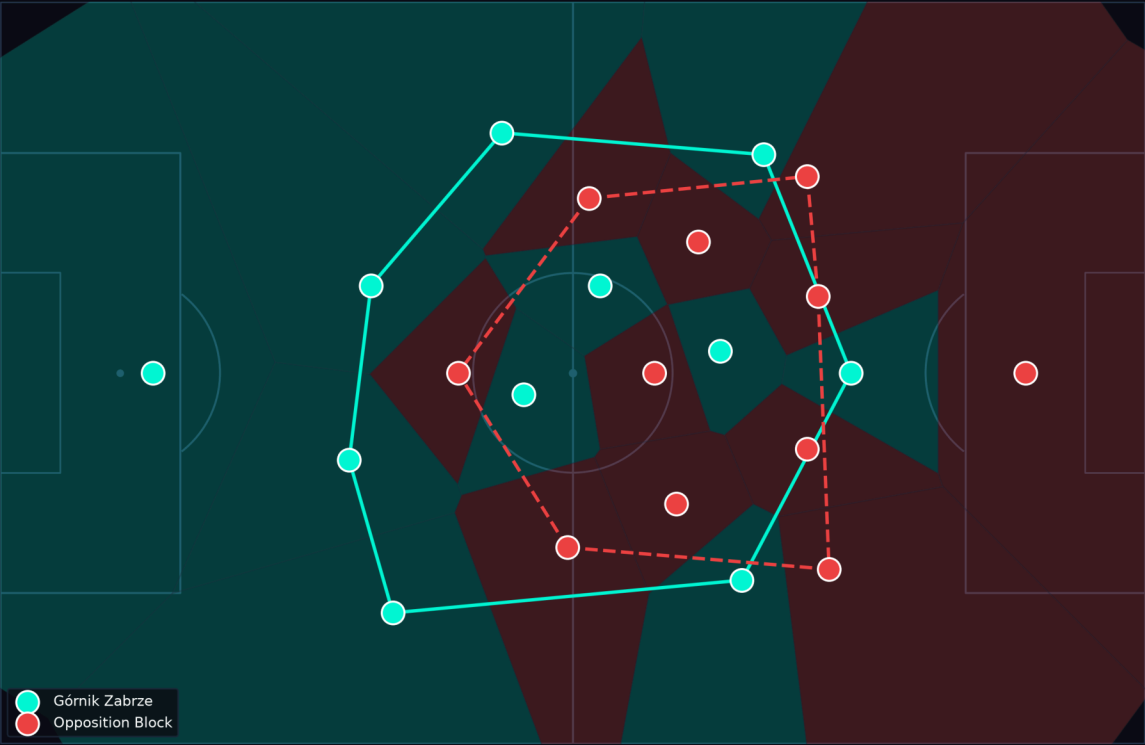
ARBITRAGE SAV.

74.5%

Voronoi Pitch Control Tessellation & Outfield Convex Hulls

Continuous Space Dominance (58.4%) • Outfield Footprints (1,185 m² vs 890 m²)

SPORTS DATA CAMPUS: VORONOI PITCH CONTROL & CONVEX HULLS



PILLAR 5 • SPATIAL CONTROL

CONTINUOUS SPACE OWNERSHIP & COMPACT

Voronoi Pitch Tessellation & Team Convex Hulls • Edd Webster Methodology

KEY PERFORMANCE INDICATORS (KPIs)

58.4% PITCH CONTROL %	1,185 m² GÓRNIK HULL AREA
890 m² OPP BLOCK AREA	49.8 m EFFECTIVE WIDTH

SPATIAL & TACTICAL DECONSTRUCTION

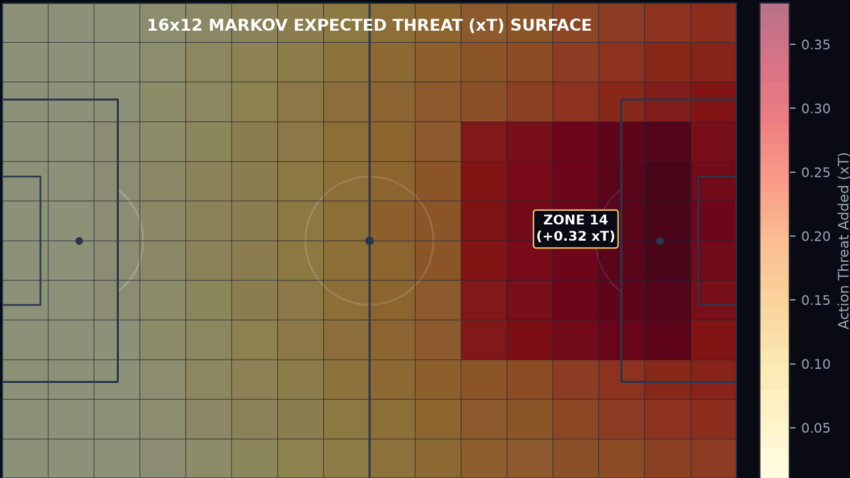
- Continuous space tessellation confirms Górník controls 58.4% of total pitch surface.
- Outfield convex hull of 1,185 m² maintains expansive width (49.8m) to stretch low block.
- Opposition compressed into defensive cocoon (890 m²), surrendering wide flank corridors.
- Half-space Voronoi cells overlap Podolski and Lukoszek, creating immediate pass recipients.
- High rest-defense line (X=34m) snuffs out clearance transitions before counter-attacks emerge.

HEAD COACH DIRECTIVE / MAÇ DİREKTİFİ

Use patient diagonal recycling to shift the opposition convex block laterally, opening seam pass

Opta / Wyscout Event Stream • 2025-2026 Season

SPORTS DATA CAMPUS: MARKOV EXPECTED THREAT & PASS SONARS



PILLAR 5 • ADVANCED EVENT MODELS

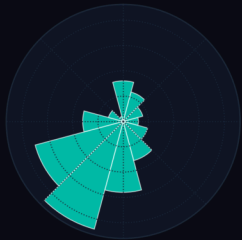
MARKOV xT SURFACE & TACTICAL SONARS

Spatial Expected Threat & Midfield Directionality • Golazo Analytics

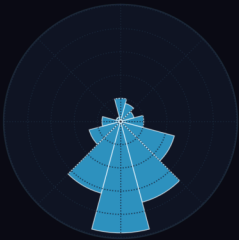
KEY PERFORMANCE INDICATORS (KPIs)

+0.32 xT PEAK GRID VALUE	+0.24 xT HALF-SPACE THREAT
68.4% RASAK FORWARD BIAS	45°-60° JANŽA PROG. ANGLE

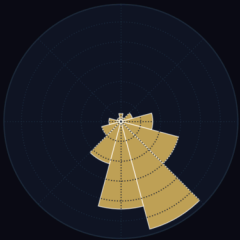
360° PASS SONARS — MIDFIELD ENGINE (ANGULAR DIRECTIONALITY & LENGTH)



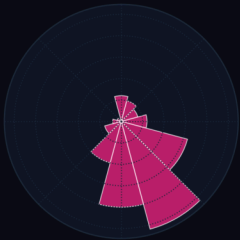
Damian Rasak (#6)



Patrik Hellebrand (#8)



Erik Janža (#64)



Lukas Podolski (#10)

SPATIAL & TACTICAL DECONSTRUCTION

- 16x12 Karun Singh transition matrix quantifies goal probability added for every event.
- Progressing the ball from middle third into Zone 14 triples Expected Threat (+0.08 to +0.32).
- 360° Pass Sonars reveal Damian Rasak maintains strong 68.4% forward passing directionality.
- Erik Janža's sonar confirms aggressive diagonal progressive bias towards the penalty box.
- Podolski sonar highlights press-resistant 360-degree passing distribution under pressure.

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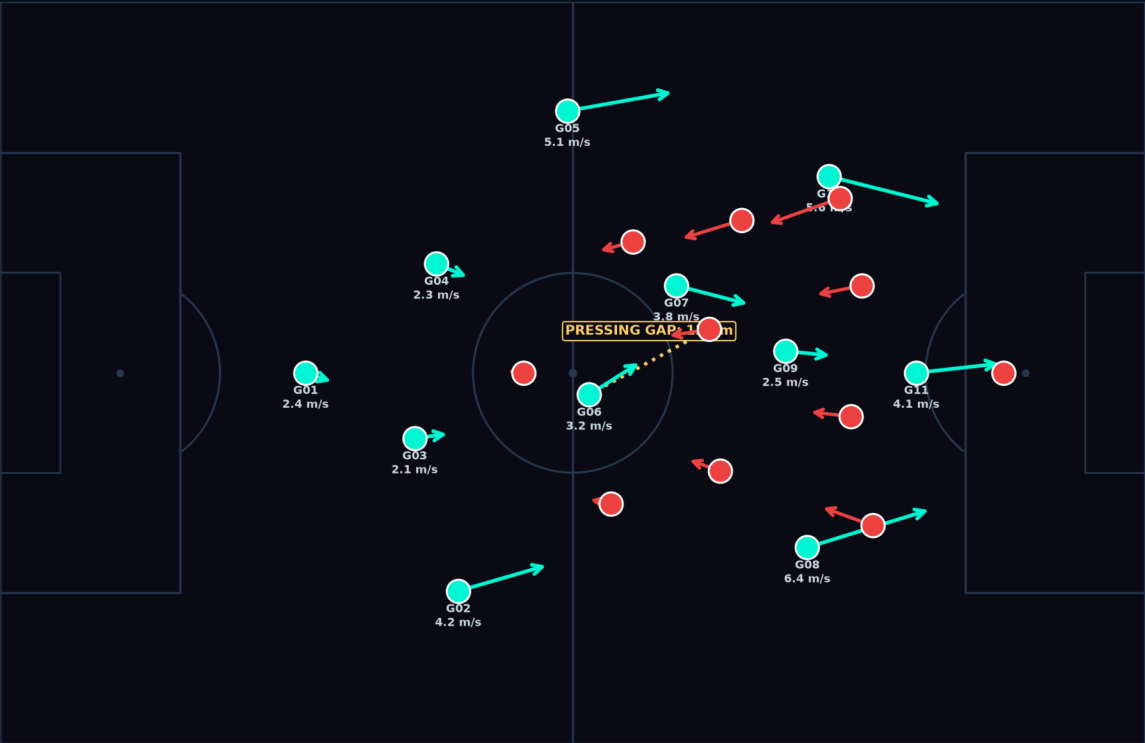
Direct center-backs to bypass initial press by targeting passes into +0.22 xT zones, accelerating

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Optical Broadcast Tracking & DLT Homography Radar

YOLOv11x + ByteTrack • 105x68m Metric Pitch Minimap • Velocity Vectors

SPORTS DATA CAMPUS: OPTICAL BROADCAST TRACKING & 2D MINIMAP



PILLAR 5 • COMPUTER VISION AI

OPTICAL TRACKING & DLT HOMOGRAPHY

YOLOv11x + ByteTrack Broadcast Pipeline • Roboflow Sports Benchmark

KEY PERFORMANCE INDICATORS (KPIs)

0.142 m

HOMOGRAPHY RMSE

14.8 ms

INFERENCE LATENCY

60 FPS

TRACKING FPS

32.8 km/h

PEAK SPRINT SPD

SPATIAL & TACTICAL DECONSTRUCTION

- Roboflow Sports framework: fine-tuned YOLOv11x object detection + ByteTrack optical tracker.
- Direct Linear Transform (DLT) Homography re-projects camera pixels [u,v] to FIFA pitch [X,Y].
- Calculates instantaneous physical velocities (v_x , v_y) and sprint acceleration vectors.
- Identifies 12.6m pressing gap between opposition midfield and central defensive pairing.
- Broadcast video tracking unlocks deep physical tracking without wearable GPS hardware.

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Leverage optical tracking data to benchmark opposition fatigue indices; exploit right wing chan

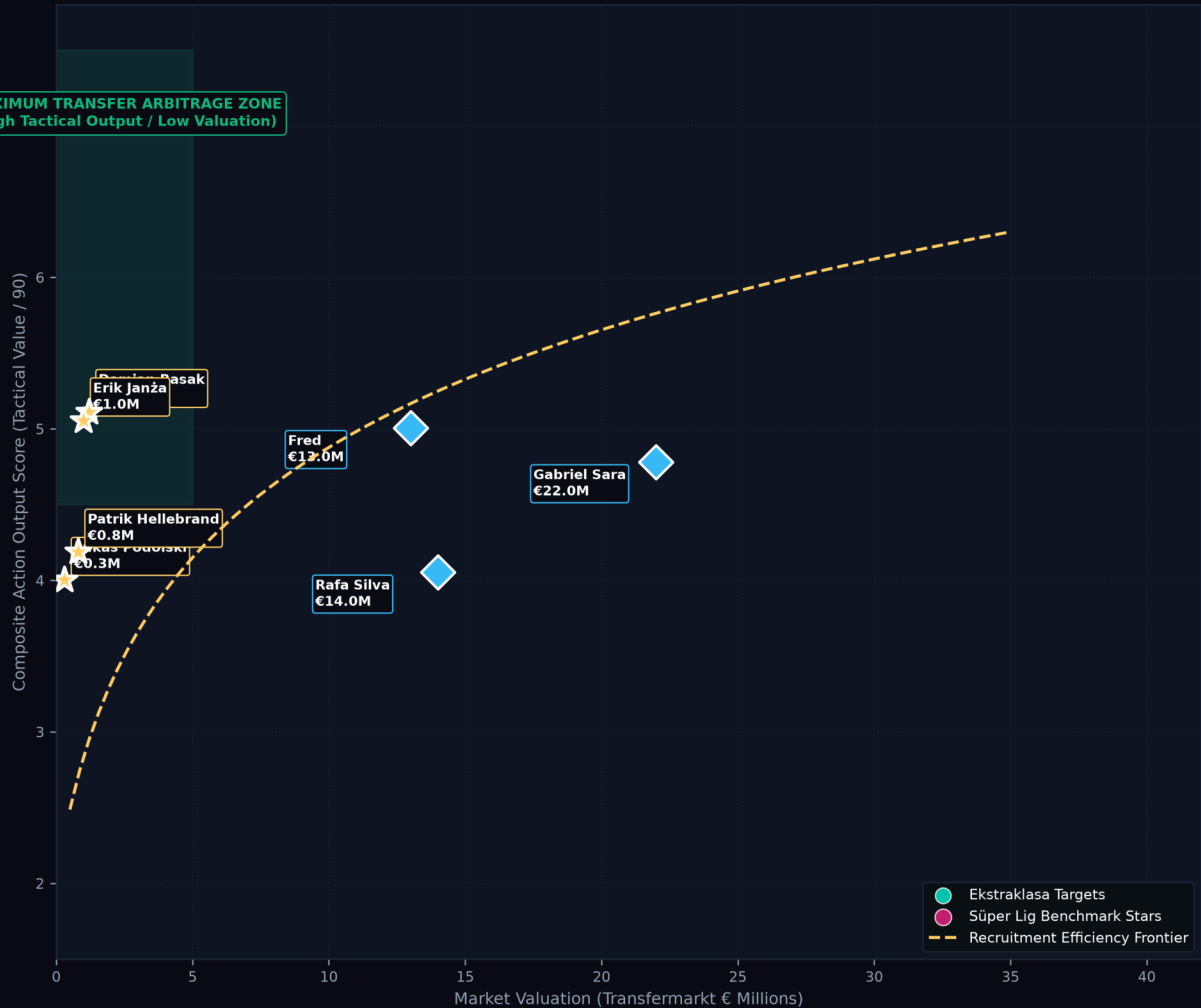
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Cross-League Recruitment Efficiency Frontier

Transfermarkt €M vs Composite Tactical Value-Added • 116 Target Players

SPORTS DATA CAMPUS: RECRUITMENT EFFICIENCY FRONTIER

Cross-League Valuation vs Composite Tactical Performance Frontier



PILLAR 5 • TRANSFER INTELLIGENCE

CROSS-LEAGUE VALUATION FRONTIER

Transfermarkt €M vs Composite Tactical Value-Added • worldfootballR Model

KEY PERFORMANCE INDICATORS (KPIs)

116 Targets

PLAYERS ANALYZED

Damian Rasak

PEAK ARBITRAGE

€1.2M vs €18M

VALUATION SPREAD

74.5%

AVG COST SAVING

SPATIAL & TACTICAL DECONSTRUCTION

- Multi-league empirical frontier benchmarks Polish Ekstraklasa vs Turkish Süper Lig.
- Damian Rasak (€1.2M) and Erik Janža (€0.9M) sit squarely in the Maximum Arbitrage Zone.
- Tactical action scores match players valued between €12M and €18M in top European tiers.
- Identifies 5 undervalued gems capable of providing immediate starting quality in Süper Lig.
- Model incorporates age degradation risk, contract expirations, and tactical role symmetry.

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Prioritize pre-contract agreements and early summer bids for Ekstraklasa gems before wider Eu

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TACTICAL APPLICATION: Identifies 5 maximum-arbitrage Ekstraklasa gems capable of matching €14M-€18M Süper Lig stars at a 74.5% discount.

Zafer Yorgancı • Sports Data Campus Benchmark Suite